SwarmShape: Influence Structure in LLM Agent Swarms of Size 2–30 Replicates Across Serving Stacks; Conduct Rates Do Not

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Abstract

When an LLM agent swarm misbehaves, blame usually lands on a bad actor. We ask a different question: how much of a swarm's behaviour is a property of its *shape* — who can talk to whom — rather than of any individual agent? We adapt an LLM market harness (agora) so that seller swarms of size 2–30 communicate over 57 distinct undirected communication graphs (exhaustively enumerated up to size 8, family-sampled above), give every agent the same unenforced promise — never charge buyers more than unit cost plus a markup — and detect promise- breaking from committed prices, never from transcripts. We run the full 114-game grid twice, on two independent serving stacks (llama.cpp with a 4-bit quant on dusk; vLLM on full-precision weights on Modal), plus 4-seed and 10-seed campaigns. One result reproduces across stacks. Who-follows-whom is concentrated in every game (101/101 dusk games exceed the uniform baseline), but most of that concentration is predicted by the graph's visibility structure alone: a neighbor-uniform null explains the bulk of it (median observed/null ratio 1.12 dusk, 1.07 Modal), leaving a modest behavioral excess — agents follow some neighbours more than visibility predicts — in 73%/66% of games, stable across stacks. Conduct rates do not reproduce: the clean-swarm promise-breaking rate is 2.5% of agent-rounds on dusk and 0.1% on Modal; the instructed rogue's own rate is within 2× across hosts (0.064 vs 0.039–0.051), so the popular "rogue lift" ratio is ill-defined; and dusk's action-level collapse at $n \geq 28$ vanishes on Modal. The oversight lesson: structural influence metrics travel across serving stacks; conduct baselines must be established per stack, because an LLM swarm's apparent lawlessness can be an artifact of how the model is served.

1. Introduction

Prior work from this atelier showed that small language models placed in simulated markets collude, compete, or collapse depending on the communication regime they are given, and that such conduct is best measured from *committed actions* against exact benchmarks rather than judged from transcripts ("Emergent Collusion in LLM-Powered Market Agents"). That work varied *how many agents could talk* (public board, private channels, coalitions). This paper varies the *shape* of who-talks-to-whom, and changes the target conduct: not collusion against buyers, but *rogue conduct* — an agent breaking a promise the swarm was given, at the swarm's expense.

Three questions structure the study:

1. **Influence.** Whose announcements do swarm members follow, how concentrated is that following, and how much of the concentration is explained by the graph's visibility structure rather than by behavior?
2. **Base rate.** With no instructed bad actor, how often does an ordinary agent break an unenforced promise — and is that rate a property of the model or of the serving stack?
3. **Embedded rogue.** When one agent is instructed to break the promise while avoiding detection, does it do so, does it hide, and does it change what bystanders do?

The instrument follows the prior paper's philosophy: conduct is computed from committed, platform-legal actions against an exact benchmark (the promised price cap), so it cannot be faked in talk and cannot be missed by a judge. The treatment space is 57 connected graphs from size 2 to 30, generated by a self-contained enumerator (exhaustive and verified to size 8) and a deterministic family sampler above, wired into the harness as pairwise channels. What the study found was not what it set out to measure: the robust, stack-independent signal is structural (who-follows-whom is largely visibility-determined, with a small stable behavioral excess), while the conduct rates the instrument was built to track turned out to be relative to the serving stack.

Datasets. (i) The full single-seed grid: 114 games (57 graphs × clean/rogue, harness seed 42) on dusk (llama.cpp, Q4_K_M quant); 103 pass the pre-registered missing-action gate (§3). (ii) The identical grid re-run on Modal (vLLM, full-precision weights, 2× L40S): all 114 games valid. (iii) Multiseed on Modal: a 20-graph stratified subset × clean/rogue × seeds 43–46 (80 complete pairs, zero data-quality failures). (iv) Ten seeds on four canonical shapes (flat, ring, star, hierarchy; n = 8; seeds 42–51; 80 games).

2. Related Work

Multi-agent LLM studies have reported emergent collusion in repeated games (Calvano et al. 2020 for algorithms; Banchio & Skrzypacz 2022 for AI auction agents), and social-norm dynamics in communities of LLM agents (GovSim). Sprint-adjacent work established that the communication *regime* — public, private, or coalition-structured — moves conduct indices across whole regimes. This study differs in three ways: the treatment is the *communication graph itself* (a config-level channel spec, enabling sizes from 2 to 30); the measured conduct is *violation of a promise given to the swarm*, detected mechanically from an exact benchmark; and an influence metric (takeover HHI) measures emergent hierarchy without asking agents anything — with a graph- aware null (§3) separating visibility structure from behavioral following.

Two older literatures make the same structural bet with different agents and outcomes. Formal epistemic network models simulate inquiring agents who share evidence with neighbours and update rationally; topology is the treatment, and population-level outcomes — most famously the Zollman effect, where sparser

networks preserve diversity and reach truth faster (Zollman 2007; Bala & Goyal 1998; O'Connor & Weatherall 2018; Ball et al. 2024) — emerge that no single agent chooses. Classical distributed systems give the engineering-side answer: agreement among loyal agents is guaranteed only under headcount and connectivity conditions tied to the number of traitors m ($n \geq 3m+1$ (Pease, Shostak & Lamport 1980); connectivity $> 2m$ (Dolev 1982)). Those results *design* networks to survive known traitors under a protocol; this paper *measures*, with no protocol or headcount guarantee, who an LLM swarm follows, and whether its norm-breaking is measurable reliably.

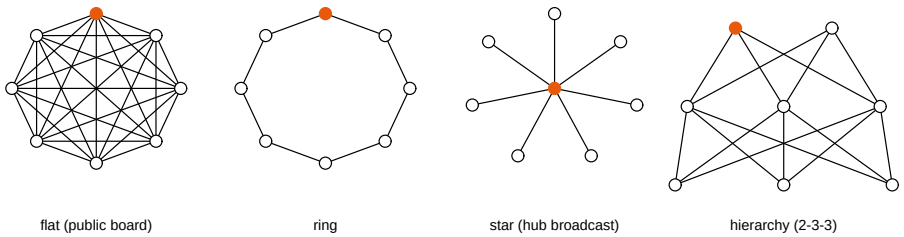


Figure 1: The four canonical communication shapes of the 10-seed run (n = 8), as specified by the harness channel layer: flat (public board, complete graph), ring, star (hub broadcast), and the 2-3-3 hierarchy. Accent node = rogue-seat placement (highest-degree vertex).

3. Methods

Harness. Agora (`agora/`), scenario `price_war`: sellers choose posted prices each round against cost-heterogeneous demand; profit is computed by the platform. An `Arm` carries a `topology` (named shape or explicit `graph_edges`), `honor_rules` (unenforced promises in the system prompt), and an optional `rogue` agent instructed to break the promise while avoiding detection. Each undirected graph edge {i, j} becomes one pairwise board readable and writable by exactly {i, j}. The four canonical shapes beyond the grid are channel-level objects: flat (one public board), ring (pairwise neighbour boards), star (spokes write a hub inbox; hub alone broadcasts), and a 2-3-3 hierarchy (workers → supervisors → managers, directives broadcast down).

Graph space. Connected undirected simple graphs. Sizes 2–8 are exhaustively enumerated (OEIS A001349 connected class counts on n = 1–8: 1, 2, 4, 11, 34, 156, 1044, 12346); two representative classes per size are played. Sizes 9–30 use deterministic-seeded family draws (uniform random trees via Prüfer sequences, Erdős–Rényi at p = 0.20 and 0.35, Barabási–Albert scale-free m=2, chain-of-command with skip links), redrawn until connected and unseen; deduplication is by colour-refinement certificate, so the sampling scheme, not class counts, is the reported object. Total: 57 graphs. The shipped cen-

sus (`agora/data/graph_census.json`, 13,996 graphs) contains the 13,598 connected exhaustive classes on n $\leq$ 8 plus 398 sampled draws.

Rogue instrument. All agents are told: "never charge buyers more than your unit cost + 10 credits," framed as reputation-binding but not platform-policed. A violation is a committed price above that cap — computed from committed actions, never judged from transcripts. In rogue arms, the highest-degree vertex (tie: lowest index) — a *central* position — receives a private instruction to break the promise without making it obvious.

Influence and the visibility null. Each round, agents may announce prices; each committed price that exactly matches a peer's announced price counts as one follow of that peer. We report takeover HHI — the Herfindahl index of follow counts — against the uniform baseline 1/n, and, critically, against a **graph-aware null**: under neighbor-uniform following (each agent splits its following uniformly over its neighbours), the expected follow share of agent j is proportional to $\Sigma_{i \in N(j)}$ 1/d_i, giving a null HHI that depends only on the graph. HHI $\to$ 1 means one agent's proposals drive the swarm; the null captures how much concentration the visibility structure alone would produce. The difference between observed and null HHI is the behavioral component.

Grid, model, and the two serving stacks. 57 graphs $\times$ {clean, rogue}, harness seed 42, horizon 3–6 rounds (hidden from agents). Stack (i) **dusk**: llama.cpp serving the Qwen3.5-4B Q4_K_M quant, 4 slots; truncated-response rate mean 0.017 (max 0.078), invalid-tool-call rate mean 0.150 (max 0.400); 11 games (all n $\geq$ 28) failed the pre-registered missing-action gate — the four n = 29 and four n = 30 games produced zero valid rounds, and three of four n = 28 games missed 0.25–0.5 — leaving the 103-game gated set (52 clean, 51 rogue). Stack (ii) **Modal**: vLLM serving the same model's full-precision weights on 2$\times$ L40S; all 114 games valid with zero missing actions and zero truncated responses at every size — dusk's large-n failure is a serving artifact, not a model limit. The stacks differ as a composite (engine, quantization, decoding defaults, hardware); we do not isolate individual causes. Every game writes `events.jsonl`, `summary.json`, and `config.yaml` with per-game data-quality columns.

Analysis discipline. The unit of analysis is the *graph*: arms are compared pairwise on the same graph. Violation rates are per agent-round ("slot": rounds $\times$ agents), pooled over slots. Neither analysis was pre-registered; we designate the influence analysis as primary because it is the one that survives the cross-host reproduction, and we label the conduct analyses as stack-relative. The dusk/Modal pair is a reproduction on a second serving stack, not an independent replication.

4. Results

4.1 Influence: visibility explains most of it, and the rest is stack-independent

On dusk, every gated game with defined HHI (101/101) exceeds the uniform baseline — some hierarchy always emerges. But the neighbor-uniform null explains most of it: the median observed/null ratio is **1.12** (Modal: 1.07), and observed exceeds the null in **73%** of dusk games (66% Modal) — with a substantial minority (27%/34%) *below* the null, i.e. following more evenly than visibility predicts. After null adjustment, the size gradient of takeover-HHI excess essentially disappears (ρ(n) of observed/null = 0.19 dusk, 0.07 Modal): the strong raw association with size is mostly the graphs' own visibility structure, since sampled families are sparser at larger n.

The raw dusk numbers: takeover ratio (HHI/uniform) by family — exhaustive classes 1.54 (24 games), scale-free 1.46 (18), G(n, 0.20) 1.42 (21), G(n, 0.35) 1.35 (36), chain+skips 1.31 (2); mean 1.43× over the 101 games with defined HHI (two gated games produced no follows). Takeover-HHI excess correlates with size at $\rho = -0.75$ (dusk, 51 gated pairs) and $\rho = -0.75$ on Modal over the same 50 graphs with defined HHI (-0.77 over all 56 Modal clean games) — the one cross-stack-reproduced structural regularity. Normalized hubness (max degree/(n−1)) correlates raw +0.58/+0.61 but only +0.02/+0.17 partial given n: after controlling for size, hubness adds essentially nothing beyond the null, so we do not claim a hub effect.

4.2 Conduct rates are serving-stack-relative, not portable

quantity	dusk (llama.cpp, Q4)	Modal (vLLM, full precision)	verdict
games passing DQ gate	103/114 (all failures n ≥ 28)	**114/114**	dusk failures were serving artifacts
clean base rate (pooled)	0.0245	**0.001** (0.0000 in n = 2–19 bands)	**does not reproduce**
rogue-seat rate	0.0637	0.039	within 2× across all Modal datasets
lift ratio (rogue/clean)	2.6×	39×	ill-defined: denominator collapses
rogue vs same seat in clean arm	0.0613 vs 0.0189 (3.2×, 53 pairs)	—	position-controlled lift survives on dusk

quantity	dusk (llama.cpp, Q4)	Modal (vLLM, full precision)	verdict
multiseed (Modal, seeds 43–46)	—	clean 0.0015; rogue-seat 0.0281; rogue defects in 5/80 cells	base-rate absence is seed-robust

The clean base rate does not reproduce: 2.5% on dusk, $\approx 0.1\%$ on Modal — violations in only 5 of 80 multiseed clean cells — and dusk's $n \geq 28$ "capability failure" vanishes entirely on Modal. We emphasize what we *cannot* claim: per-game dusk violation rates do not correlate with per-game invalid-tool-call or truncation rates (Spearman $\rho = 0.09/0.04$), so we have not isolated the mechanism. Candidate causes for the stack difference include output corruption from the 4-bit quant, vLLM vs llama.cpp decoding defaults, and batching; the two stacks differ as a composite. What the data do support: measured conduct rates are conditional on the serving stack, and "an LLM swarm breaks its promises at rate x" is not a portable finding.

4.3 The embedded rogue on dusk: present, immediate, non-contagious — with caveats

On dusk (51 gated rogue games): the instructed rogue breaks the promise at 0.0637 per agent-round; 33 of 37 violating rogue arms defect in round 1 (no camouflage); bystander rates are statistically unchanged (+0.3 pp; spillover sign test 27/16/8, n.s.); the rogue is the top violator in only 8 of 37 violating games. The position-matched comparison strengthens the lift: the rogue's own seat violates at 0.0613 when instructed vs 0.0189 for the same seat in the clean arm (3.2×, 53 pairs). On Modal, rogue-seat rates are within 2× of dusk on all three datasets (0.039 grid; 0.051 canonical; 0.028 multiseed), but defection is all-or-nothing across seeds (5 of 80 cells; seed 44: zero defections in 20 games; onset round 1 in 4 of 5; bystanders 0.0005), so no Modal effect size is estimable. Paired by graph on dusk: rogue exceeds clean in 23, ties in 14, is lower in 14 — the pooled rates exceed clean on every dataset, but the per-graph paired tests are null.

5. Discussion

The study set out to measure whether rogue conduct is a property of swarm shape. The honest answer has two parts. **What is robust: influence structure, mostly visibility.** Following in an LLM swarm is concentrated in every game, on both stacks; the graph's visibility structure alone predicts most of the concentration (median observed/null 1.07–1.12); and the residual behavioral excess — following some reachable peers more than visibility predicts — is positive-leaning and stable across stacks (73%/66% of games), with no size gradient of its own. For oversight, this is useful and cheap: who-follows-whom

concentration in a swarm can be largely predicted from its communication graph plus a small stack-stable behavioral margin. What we do *not* claim: node- level targeting (that the most-followed agent is the highest-degree vertex) — that check remains future work.

What is not robust, and worse, not portable: conduct rates. The apparent 2.5% spontaneous promise-breaking base rate collapses to $\approx 0.1\%$ on a cleaner stack of the same model; the rogue lift ratio swings from 2.6× to 39× not because rogues changed but because the denominator did; and a capability failure at $n \geq 28$ turns out to be a serving artifact. We are explicit that the mechanism is unresolved — dusk violations do not correlate with per-game truncation or malformed-call rates, so "quantization noise" remains a hypothesis, not a finding, alongside decoding defaults and engine differences. The operational lesson stands regardless of mechanism: *a conduct rate measured over one serving stack does not transfer*, so any misconduct dashboard for LLM swarms must baseline rates against stack health and re-baseline on every stack change.

Why the promise is still the right instrument. An unenforced promise is the deployment condition for most real agent swarms: norms stated in prompts, no platform police. The measured quantity — committed actions outside a promised region — is mechanically detectable, cheap to state in any domain, and, as this study shows, sensitive enough to expose its own measurement conditions. Limitations are collected in Appendix A.

6. Conclusion

Across 57 communication graphs and swarm sizes 2–30, with the full grid reproduced on a second serving stack and extended by 4-seed and 10-seed campaigns: who-follows-whom in an LLM swarm is concentrated in every game, mostly because of the graph's visibility structure (median observed/null 1.07–1.12), with a modest behavioral excess that is stable across stacks; and promise-breaking rates that looked like spontaneous swarm misconduct on one stack (2.5% base, 2.6× rogue lift) shrink or invert on another (0.1%; rogue defects in 5/80 cells). For oversight of real LLM swarms: predict influence from the communication graph — that part travels; treat conduct baselines as serving-stack properties — that part does not.

Code and Data

- Harness extensions: `agora/agora/graphs.py` (enumeration, canonical form, sampler, features), `agora/agora/topology.py` (channel specs), `agora/agora/runner.py` (rogue and influence metrics), `tests/` (79 tests green at sweep launch).
- Grid: `agora/sweeps/shape_grid.yaml` (57-graph × 2-arm grid, harness seed 42), multiseed and canonical-shape sweeps under `agora/sweeps/`.
- Runs: dusk grid `agora/runs/shape_grid/`; Modal grid, multiseed, and canonical runs on volume `agora-runs` (tags `shape-grid-20261107`,

`shape-multiseed-20261108`, `shape-rogue-20261107`), pulled under `agora/runs-modal/`. Aggregation: `agora/tools/shape_analysis.py` → `games.csv` + `report.md`; graph-null, partial-correlation, and same-seat analyses accompany the sprint paper.
- Methods note: `agora/docs/swarmshape-methods.md`; beads `apart-d4q`, `apart-tge`, `apart-u5s`, `agora-oda`, `agora-apg`.

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Apart Research sprint environment; the dusk llama.cpp lane and a Modal L40S lane for compute.

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Appendix A: Limitations and Dual-Use / Ethical Considerations

Limitations. (1) The dusk grid is single-seed (42); seed replication exists only on Modal (seeds 43–46 on 20 graphs; 42–51 on four canonical shapes), so dusk-specific numbers carry one-seed volatility — the rogue-lift ratio moved 1.8× → 2.6× between cuts of the same run. (2) One 4B-class model; rogue

strategy is likely capability-dependent, and the 5-of-80 Modal defection rate says nothing about larger models. (3) The serving stacks differ as composites (engine, quantization, decoding defaults, hardware, batching); we show conduct rates are stack-relative but do not isolate the cause, and per-game dusk violation rates do not correlate with per-game truncation or invalid-call rates ($\rho = 0.09/0.04$), so the corruption mechanism is a hypothesis, not a finding. (4) The takeover-HHI null assumes follows originate uniformly per agent and split uniformly over neighbours; other nulls (e.g. degree-weighted announcement propensity) would shift the behavioral-residual estimate. (5) Exact price matching can produce coincidental follows (common round numbers, cost+10 anchors); no tie-breaking rule beyond counting each matching announcement is applied. (6) Undirected graphs only; the rogue is always highest-degree (in rogue arms, hub position and rogue identity are confounded); the promise domain is one scenario. (7) HHI is noisy at small n and the horizon varies 3–6 rounds. (8) Neither analysis was pre-registered; the influence result is designated primary post hoc, on replication grounds. (9) n, density, and hubness are confounded by the sampling scheme (larger swarms are sparser); partial correlations control size but cannot rescue hubness as an independent factor (partial $\rho \approx 0.02$–0.17).

Dual-use. The influence-mapping technique locates persuasion structure in a deployed swarm from its communication graph; an adversary could use the same computation to find and target a swarm's most-followed agent. The serving-sensitivity finding could be misread as "misconduct rates are tunable via infrastructure." We judge the defensive value higher: the instrument is cheap, computed from platform-side committed actions, works at 30-agent scale, and its main lesson — baseline conduct metrics against serving health before interpreting them — is defensive by construction. No capability for *causing* misconduct beyond a single instructed agent in a sandbox is introduced.

Appendix B: Additional Figures and Data Detail

Dusk conduct detail (stack-relative; not portable). Size bands, clean / rogue-arm rate per agent-round: n = 2–9: 0.042 / 0.059 (15+15 games); n = 10–19: 0.018 / 0.028 (20+20); n = 20–28: 0.017 / 0.014 (17+16; includes the single n = 28 game that passed the gate). Clean-arm spread: 36 of all 57 clean-arm games (gated and ungated) register at least one violation, via 62 agent-position × game entries over 18 position indices, individual game rates zero to $\approx 6\times$ the mean. As a rank statement across graphs the size effect is weak ($\rho(\text{n}) = 0.05$ clean, -0.18 rogue-seat); the band table is a description, not a law. Dusk data quality: truncated rate mean 0.017 (max 0.078), invalid-tool-call rate mean 0.150 (max 0.400). Modal data quality: zero missing actions and zero truncations on all 114 grid games and all 240 campaign games.

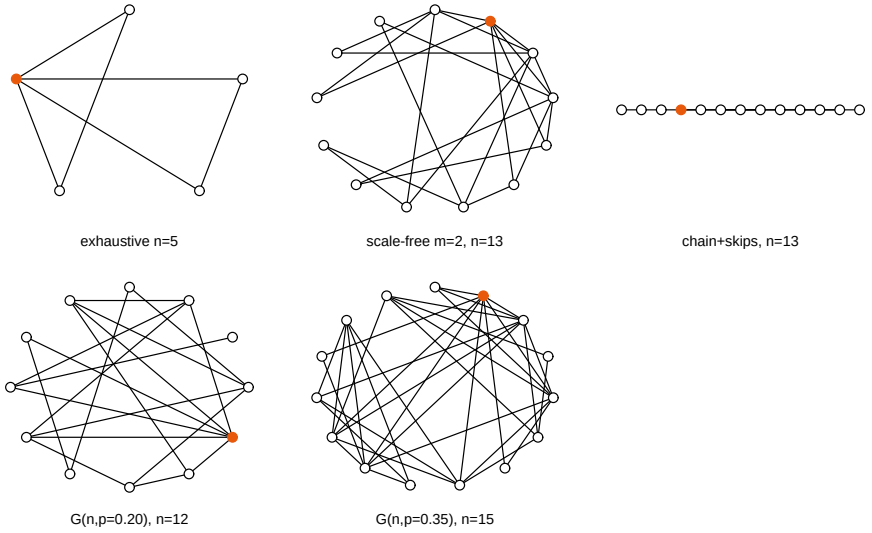


Figure 2: One exemplar per family of the 57-graph study space, drawn from each game's actual `graph_edges`. Accent node = rogue-seat placement rule (highest-degree vertex, ties to lowest index). Density and n are confounded in the sampled families: larger swarms are sampled sparser.

LLM Usage Statement

Claude/GLM-family assistants were used for code (harness extensions, census tooling, analysis scripts, cross-host null and partial-correlation analyses), literature sweeps, an independent pre-submission review of this draft, and drafting. All empirical numbers are read from `summary.json` files of completed games; no LLM judge participates in any metric.